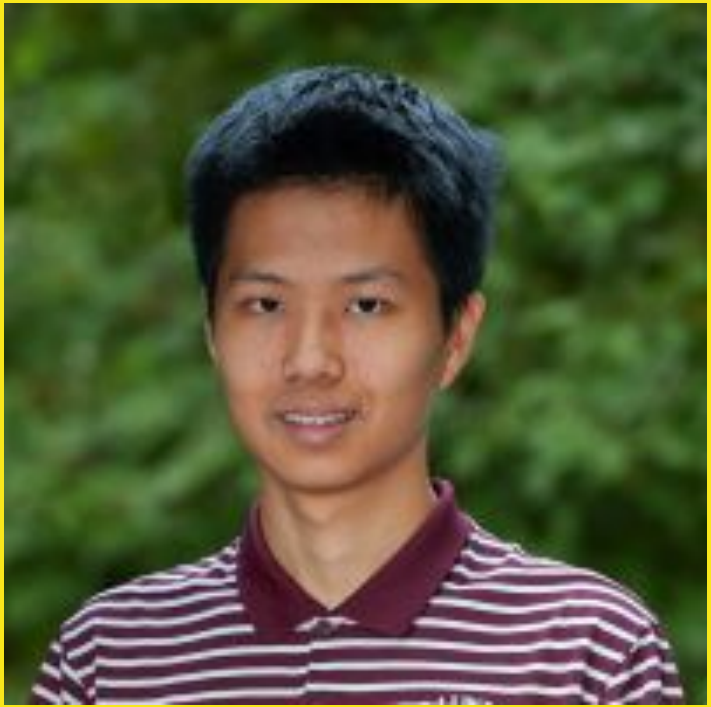
Welcome to the Artificial Intelligence Workshop (Week 4)

Let's Learn How Computers Learn from Information

About Me



Name: Peikang Hu

Current Education: Rising senior at St. Luke's School, New Canaan, Connecticut

Experiences in coding(Python, C++, Java..) and machine learning

Icebreaker

How Many AI Tools Have You Used?

Raise your hand if you've used...

- □ ChatGPT
- □ Siri or Alexa
- □ Google Translate
- □ YouTube recommendations
- □ TikTok or Instagram
- □ Face ID on a phone

Did you know?

You've probably been using AI almost every day!



Workshop Overview

Week 1 – What is Artificial Intelligence (AI)?

Goal: Spark curiosity and understand what AI is.

Week 2 – How Computers Learn

Goal: Introduce linear regression and its application.

Week 3 – A Continuation of Week 2 BUT with more Fun!

Goal: Introduce more advanced AI models.

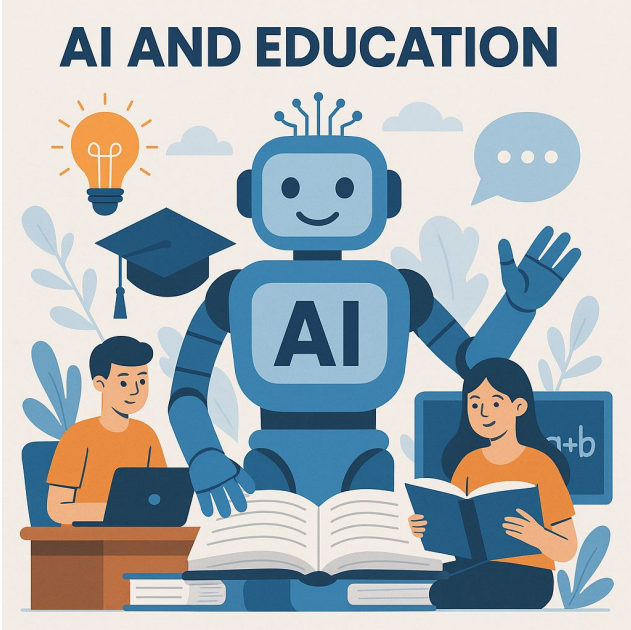
Week 4 – Prompt Engineering and Neural Network

Goal: Introduce prompt engineering and neural networks

Agenda

- Introduce Prompt Engineering
- Review the basics of Machine Learning
- Introduce neural networks and its applications

Prompting AI for Research & Homework



How to Get Better Answers from AI

AI can help you:

- Understand difficult concepts
- Brainstorm ideas
- Find different ways to solve a problem
- Improve your writing
- Study for tests
- Explore research topics

But the quality of the answer depends heavily on the quality of your prompt.

What Is a Prompt?



A **prompt** is the instruction or question you give an AI model.

Weak prompt:

> Explain photosynthesis.

Better prompt:

> Explain photosynthesis to me as if I am a 7th-grade student. Use a simple analogy and then give me 3 questions to check my understanding.

Key idea:

The more useful information you give the AI, the more useful its response can be.

What Makes a Prompt a Good Prompt?

The 5 Parts of a Good Prompt

A strong prompt often includes:

1. **Role** — Who should the AI act as?
2. **Context** — What does the AI need to know?
3. **Task** — What do you want it to do?
4. **Constraints** — What rules should it follow?
5. **Format** — How should the answer look?

Part 1: Role

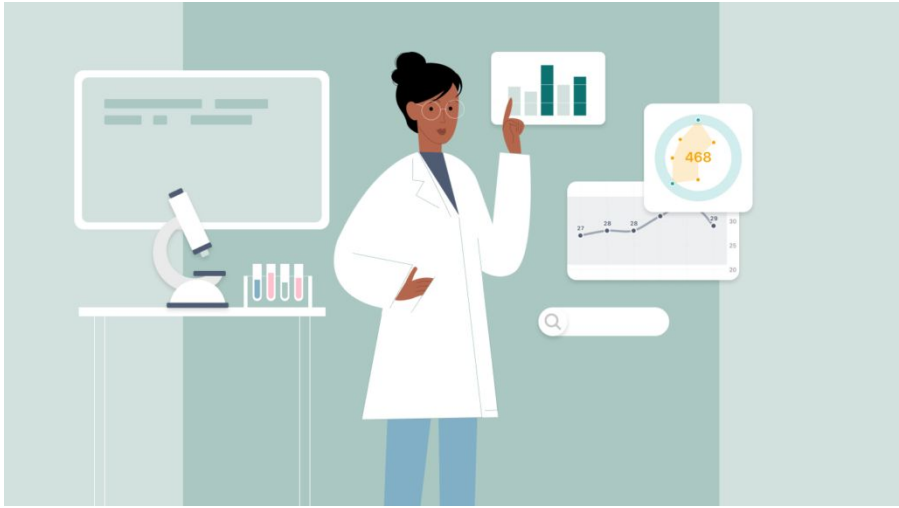
Tell AI who you want it to be.

Giving AI a role helps establish the perspective and type of response you want.

Example:

Bad: Explain calculus.

Good: Act as a high school calculus tutor.



You can use roles like:

- Math tutor
- Research assistant
- Writing coach
- Debate coach
- Programming teacher
- Science teacher

Why it helps:
The same question can receive very different answers depending on whether you ask for a tutor, researcher, editor, or expert.

Part 2: Context

Give AI the information it needs about you and the situation.

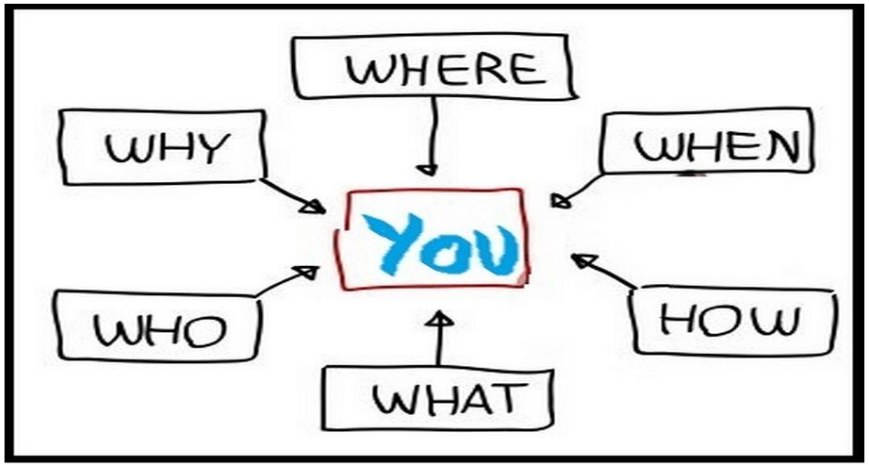
Example:

Bad: Explain mitosis.

Good: I'm a 9th-grade student taking biology, and we're currently learning about cell division. I understand basic cell structures but I'm confused about how mitosis works.

Useful context can include:

- Your grade level
- What you're studying
- What you already understand
- What you're confused about
- Your assignment requirements
- What you have already tried



Think:

"What does the AI need to know before answering me?"

Part 3: Task



Clearly tell AI what you want it to do.

Be specific about the action you want.

Compare:

Bad: Tell me about World War II.

Good: Explain the three main causes of World War II and how each contributed to the beginning of the war.

Useful task verbs:

Explain • Compare • Analyze • Summarize • Critique • Brainstorm • Quiz • Solve • Check • Rewrite

The more specific the task, the less likely you are to get a vague answer.

Part 4: Constraints

Tell AI what it should and shouldn't do.

Constraints control the **boundaries** of the response.

Examples:

Use language appropriate for a 9th-grade student.

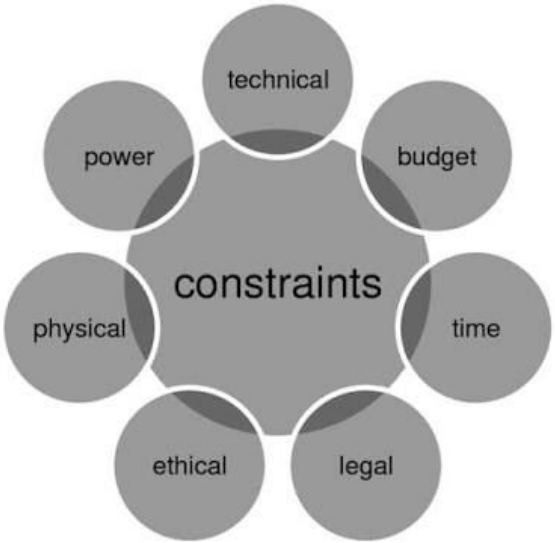
Keep the answer under 300 words.

Use only information from the article I provided.

Don't give me the answer immediately.

Give me one hint at a time.

Don't use complicated mathematical notation.



Part 5: Format

Tell AI what you want the answer to look like.

The same information can be presented in many different ways.

Examples:

Put the information in a table.

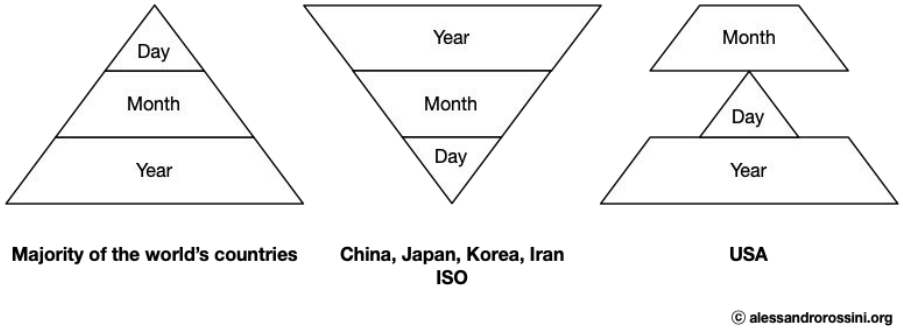
Give me 5 bullet points.

Explain it in numbered steps.

Create a timeline.

Give me a short paragraph followed by 3 examples.

Quiz me with multiple-choice questions.



Part 6: Putting It All Together

Role + Context + Task + Constraints + Format

Role: Act as my high school biology tutor.
Context: I'm a 10th-grade student studying genetics, and I understand Punnett squares but struggle with probability.
Task: Help me understand how probability is used in genetic inheritance.
Constraints: Use simple language and don't give me the answer to the practice problem immediately.
Format: Explain the concept, give me one example, then give me a practice question.

The result:

A much more **specific, useful, and personalized** response than simply asking:

"Help me with genetics."



The Socratic Approach: Make AI Ask, Not Answer

Instead of asking AI to give you the answer…

Use AI as a **Socratic assistive tutor** that guides you to the answer through questions.

Prompt:

> Act as my Socratic tutor. Do not give me the answer immediately. Ask me one question at a time to help me solve the problem myself. If I make a mistake, point it out and give me a hint rather than solving it for me.



The Socratic Approach: Make AI Ask, Not Answer

Example

Student:

Why did the American Revolution happen?

AI:

What do you already know about the relationship between the American colonies and Britain before the Revolution?

Student:

Britain was taxing the colonies.

AI:

Good. Why might the colonists have considered those taxes unfair?

Homework Help: Don't Just Ask for the Answer

Instead of:

> Solve this problem: 3x + 7 = 22

Try:

> Help me solve 3x + 7 = 22. Don't give me the answer immediately. Ask me what I think the first step should be, and guide me if I get stuck.

Why is this better?

AI becomes an assistance to your learning process, rather than an answer machine.

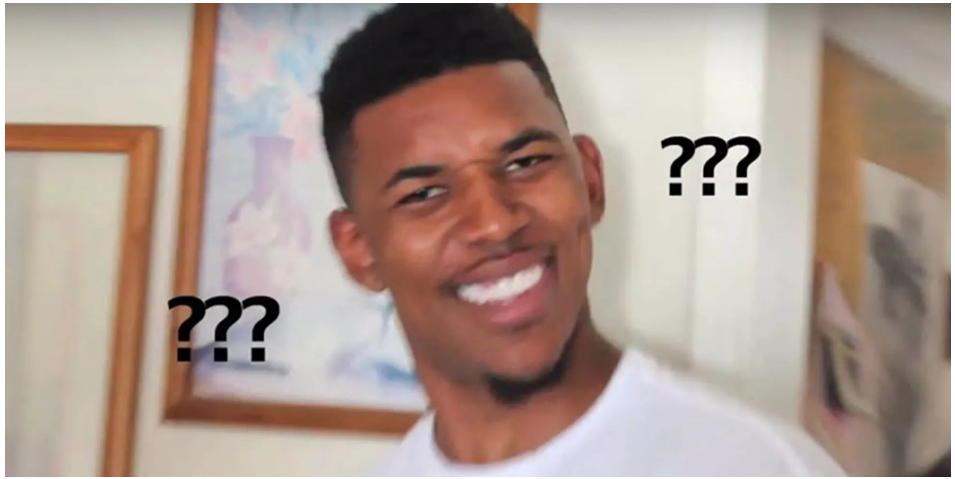
You are actually practicing the skill instead of simply copying the answer.

When You DON'T Understand an Explanation

AI's first explanation might not work for you.

Instead of giving up, **change the prompt.**

Try:



Explain this using a real-world example.

Explain it as if I'm 10 years old.

Use an analogy involving sports.

Explain the same idea without using technical vocabulary.

Give me a visual description of how this works.

AI for Research: Start With Questions



Don't immediately ask:

> Give me information about climate change.

Instead, ask AI to help you **narrow your topic**.

Example:

> I'm interested in climate change for a school research project. Give me 10 specific research questions that are narrow enough for a 5-page paper. For each question, explain why it would be interesting to investigate.

This helps turn a huge topic into a manageable research question.

AI Can Help You Brainstorm Research

Example Prompt:

> I'm researching whether social media affects teenagers' academic performance.
> Give me:
>
> - 5 possible hypotheses
> - 5 variables I could investigate
> - 3 possible explanations for the relationship
> - 3 alternative perspectives I should consider

AI can help you identify:

Questions → Hypotheses → Variables → Arguments → Counterarguments

But **you still need to evaluate the ideas yourself.**

Use AI to Critique Your Work

AI doesn't only have to generate content.

You can give it **your own work** and ask for feedback.

Prompt:

> Here is my history essay: [paste essay]
> Act as a teacher reviewing my draft. Identify:
>
> 1. My strongest argument
> 2. The weakest part of my reasoning
> 3. Any unsupported claims
> 4. Places where my evidence could be stronger
> 5. Three specific improvements I should make

AI for Research ≠ AI as Your Source

AI can help you **find and understand information**, but don't automatically treat its response as a reliable source.

AI can:

- Make factual mistakes
- Misinterpret information
- Give outdated information
- Sometimes invent citations

Better workflow:

AI → Find ideas → Find sources → Read sources → Verify claims → Cite sources

For important research, go back to the **original source**.

Ask AI to Help You Evaluate Sources

Instead of:

> Is this source good?

Try:

> Here is a research article: [article/text/link]. Help me evaluate it. Identify the authors, publication date, evidence used, methodology, potential limitations, and whether the conclusions are supported by the evidence.

Remember:

AI can help you **analyze** a source.

You should still make the final judgment.

Final Challenge: Improve This Prompt

Original:

> Help me with my history homework in the American Revolution

Challenge:

Rewrite it using:

Role + Context + Task + Constraints + Format

Final Challenge: Improve This Prompt

Act as a high school history tutor. I am studying the causes of the American Revolution. Help me understand the three most important causes. Explain each one with an example, then quiz me with 5 questions. Don't give me the answers until I try.


Neural Networks!!!

Review: Machine Learning

What Is Machine Learning?

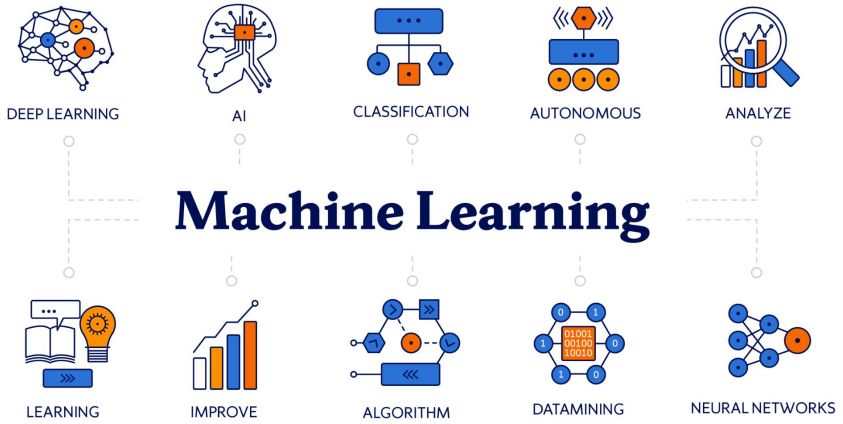
Machine Learning = Teaching computers to learn from examples instead of giving every rule.

Traditional Programming:

Human writes rules
Computer follows rules

Machine Learning:

Give examples
Computer finds patterns



Syracuse University
School of Information Studies

Three Types of Machine Learning

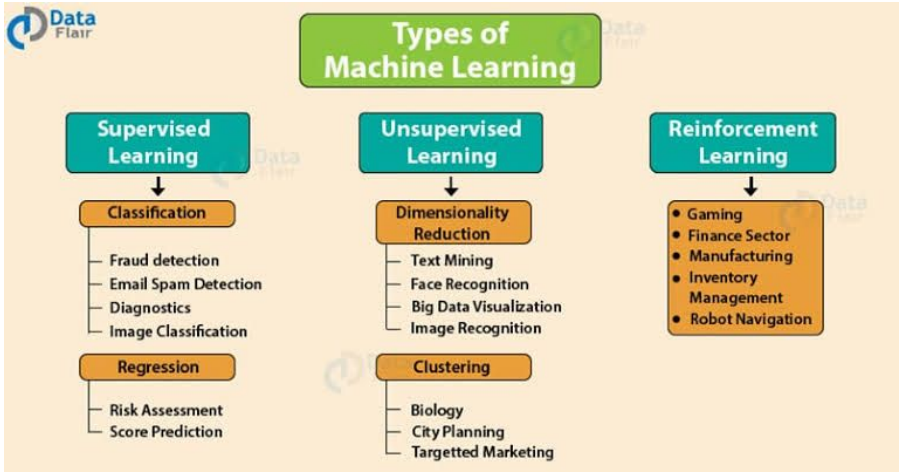
Supervised Learning

Data has **inputs AND correct answers.**

Example: Predicting Exam Scores

Imagine we collect data from several students:

- A student who studied **2 hours** scored **65**.
- A student who studied **4 hours** scored **80**.
- A student who studied **6 hours** scored **92**.
- A student who studied **8 hours** scored **97**.



Using many examples like these, a machine learning model learns the relationship between **hours studied** and **exam score** so it can predict the score for a new student.

Three Types of Machine Learning

Unsupervised Learning

Only inputs.

No labels.

Example:
Group customers based on shopping habits.

Output:
Different customer clusters.



Three Types of Machine Learning

Reinforcement Learning

An agent learns through rewards and penalties.

Examples:

- Chess
- Robotics
- Video games



Two Types of Supervised Learning

Classification

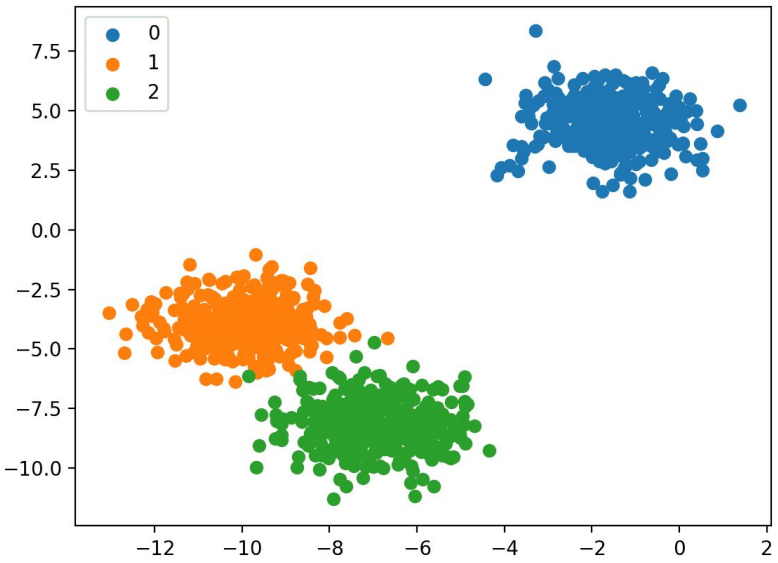
Predict categories.

Examples

- Spam or not spam
- Dog or cat
- Fraud or legitimate

Output:

Discrete labels



Two Types of Supervised Learning

Regression

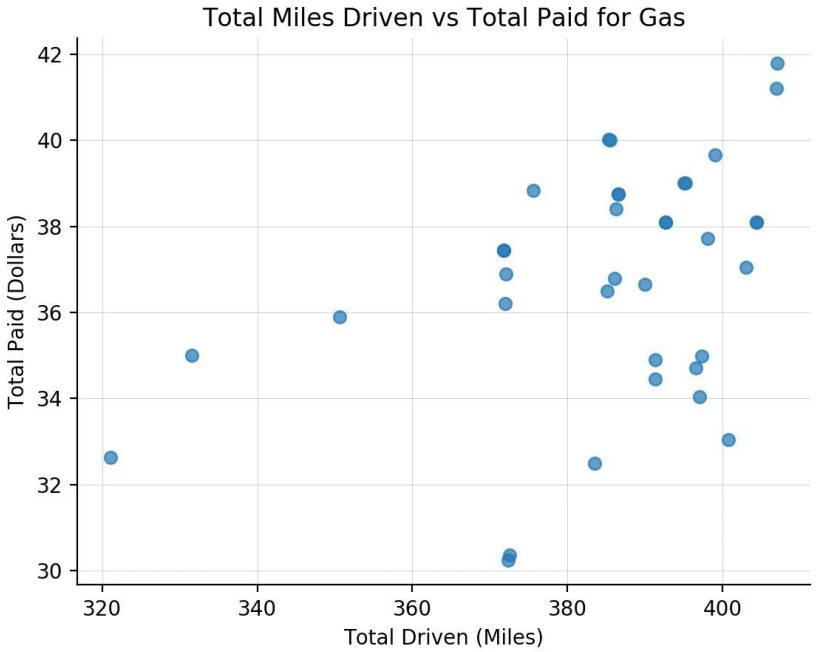
Predict numbers.

Examples

- House prices
- Temperature
- Stock prices
- Salary

Output:

Continuous values



What Is a Neural Network?

A **neural network** is a type of AI that learns patterns from data.

Instead of being given every rule, we give it examples and let it **learn the rules itself**.

Example:
Give a neural network thousands of pictures of cats and dogs.

Over time, it can learn patterns that help it decide:

🐱 **Cat** or 🐶 **Dog**

Inspired by the Brain

Neural networks are loosely inspired by how the human brain works.

The basic building blocks are called **neurons**.

Each neuron:

- Receives information
- Processes the information
- Passes information to the next neuron

In an AI model, these are mathematical calculations rather than real biological neurons.

How Does a Neural Network Learn?

A neural network starts with **random values**.

It makes a prediction → compares the prediction to the correct answer → adjusts itself.

Example:

Input: Picture of a dog
Prediction: Cat ❌
Correct answer: Dog ✅

The network adjusts its internal values so that next time it is more likely to predict **dog**.

This process happens **over and over again** with lots of examples.

Layers of a Neural Network

A simple neural network has three main types of layers:

Input Layer → Hidden Layers → Output Layer

For example, predicting whether a student will pass an exam:

Input Layers:

- Study hours
- Attendance
- Homework completion

Hidden layers:
The network finds patterns in the information.

Output Layers:

Pass: 87%

The hidden layers are where much of the pattern-finding happens.

Why Are Neural Networks Powerful?

Neural networks can learn **very complicated patterns**.

They are used for:

🖼️ **Image recognition**
💬 **Language and chatbots**
🎵 **Speech recognition**
🚗 **Self-driving systems**
📈 **Prediction and classification**

Main idea:

Traditional programming:
Human gives the computer the rules.

Neural networks:
Human gives the computer data, and the model **learns patterns from the data**.


Thank You Very Much for Coming to Today's Session!